

Adversarial Cooperation

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Abstract—Adversarial Cooperation investigates cryptographic protocols for classical and modern problems in game theory without a trusted referee, and the hardware that parties would use to execute them. The aim is to enable cooperation between adversarial parties while preserving private information and the strategic advantages that depend on it. Each construction will define what must be learned to reach an agreed result and seek to reveal nothing further under explicit assumptions. We seek resources and research support to develop, analyze, implement, and demonstrate these protocols and their hardware.

Index Terms—Applied cryptography, cooperation, zero knowledge, multiparty computation, hardware.

I. THE COOPERATION PROBLEM

The central question is: *how can we construct and demonstrate cryptographic protocols that let adversarial parties cooperate without a referee, while retaining the private information on which their strategic advantage depends?* A party should be able to take part in a common game or verify an agreed result without handing its cards, strategy, or other secrets to an opponent or a trusted intermediary.

Author’s intuition: “Playing poker without revealing the cards, or playing chess without moving the pieces.” In poker, a question is whether players can verify a winning hand without disclosing its cards.

For chess, the intuition is that visible play may work against the parties’ interests by exposing information about their strategies. If both parties fix deterministic move-selection algorithms, the proposed protocol would privately compute the outcome of their matchup from an agreed initial position. On completion, it would reveal only win, loss, or draw, seeking to hide the algorithms and move sequence beyond what the result implies. The agreed rules must specify draws and how illegal moves or exceeded computation limits affect the result. The computation may simulate play internally; this proposal does not assume a shortcut to determining the outcome.

The privacy objective is to reveal no additional private information beyond the agreed outputs and public information, under the chosen adversary model. The parties still exchange protocol messages, and the result itself may reveal information. Preserving a strategic advantage therefore requires identifying the secret it depends on and examining what the output, repeated queries, and subsequent play disclose. Execution costs and aborts must also be considered.

The program builds on commitments, zero-knowledge proofs, and multiparty computation: established ways to fix private choices, prove predicates, and evaluate joint functions under specified assumptions [1]–[3]. Cryptographic replacement of a mediator in strategic games is also prior work [4].

This proposal claims no new primitive or general solution to cooperation. It proposes carefully specified case studies that connect these foundations to the decisions parties face, and follow the resulting requirements into software and hardware. Whether any particular composition constitutes a research contribution must be established case by case.

II. SCOPE, ASSUMPTIONS, AND REQUIREMENTS

Here, *without a trusted authority* means that the selected task requires no central party entrusted with all inputs or unilateral authority over its result. It does not mean absence of assumptions. Each case study must state its authentication mechanism, channel model, randomness requirements, setup resources, corruption threshold, and reliance on local devices. A setup that introduces a trusted dealer must be identified as changing the objective.

The proposed starting model has two parties and authenticated channels, with at most one party statically corrupted by a probabilistic polynomial-time adversary. The adversary can stop or delay communication. Each construction must specify any additional channel or setup requirements and how they are met. Computational privacy and correctness against malicious deviation are initial research targets under these assumptions; they remain to be established for the proposed implementations.

For each task, an ideal functionality will specify inputs, outputs, allowed leakage, and the intended behavior of output release. Each party would be required to assess the algorithm’s privacy and correctness under the stated adversary model and conclude that the evidence justifies its use. It would then fabricate its own trusted hardware from an inspectable design. Trust would concern correct execution and protection of that party’s secrets, with fabrication and physical-access assumptions stated. Any reliance on a peer’s hardware would require separate justification.

Preventing unilateral early learning followed by strategic withdrawal is a research objective. For each selected task, we seek a mechanism that prevents one party from obtaining the protected result while withholding the messages needed for the other party’s agreed output. The analysis must establish what protection holds when a participant stops or delays communication and which communication and hardware assumptions it requires. This output-release mechanism remains to be constructed and demonstrated.

Correct evaluation does not establish input truthfulness or beneficial incentives. Any assumption of rational behavior requires an explicit utility model.

III. A SMALL, FALSIFIABLE STARTING POINT

The current technical anchor is a finite-game question: can a party demonstrate that a committed, deterministic Tic-Tac-Toe policy never loses from the empty board, for a fixed role, without disclosing the policy? The predicate quantifies over every legal opposing continuation. It does not assert optimal play at every state, ownership of the strategy, or adherence to it in later play. Tic-Tac-Toe is a bounded testbed with a checkable state space; its policies are not presumed to be valuable secrets. It tests whether a claim about a private decision rule can be carried consistently through a mathematical relation and an executable representation before considering more consequential decisions.

The intended proof relation illustrates the composition issue. Let p contain the game version, role, initial state, and session context; let s be a canonically encoded policy and r commitment randomness. For public commitment c , the proposed relation is

$$R((p, c); (s, r)) = [c = \text{Com}(p, s; r)] \wedge V(p, s), \quad (1)$$

where V checks policy validity and non-loss under the fixed game model. A prospective zero-knowledge argument of knowledge must bind both checks to the *same* hidden policy. Separate checks on unrelated witnesses would not establish this statement. Neither the relation nor a commitment alone establishes when a policy was fixed; chronology requires an authenticated ordering mechanism if a later application needs it.

The working repository contains a public policy checker, a fixed directed acyclic graph evaluator, Boolean circuits for the policy predicate, and disclosed commitment-opening checks. A Rock–Paper–Scissors demonstration implements commit–reveal in participant-local state machines, with adversarial tests that exhibit selective abort. The present artifacts are educational: no private policy proof, distributed deployment, or hardware implementation is established. Differential and negative tests check executable behavior; they do not prove cryptographic hardness or equivalence for every policy. The next step is to select and jointly justify a compatible commitment profile and proof representation for the complete relation, then measure its actual cost and leakage boundary.

IV. FROM SPECIFICATION TO HARDWARE

The longer-term agenda follows selected components from a formal specification through explicit syntax or circuit representations and C reference code to synthesizable register-transfer-level descriptions (Verilog or VHDL), then FPGA evaluation. High-level synthesis is one possible route. Behavior-preserving synthesis already has research foundations [5]; this project does not yet supply such a toolchain or proof of translation correctness.

Each translation must state what it preserves: input encoding, arithmetic, state transitions, outputs, and error behavior.

Proof-oriented circuits and hardware circuits may need different representations and cost models; a small proof circuit is not automatically an efficient hardware design. Evaluation will report proof and verification time, communication, and memory where relevant, alongside hardware area, latency, throughput, and energy under a documented workload. Comparisons will use the same functional task and disclose platform and tool versions.

Functional correspondence alone does not establish confidentiality against timing, power, electromagnetic observations, or injected faults. These need separate models and evaluation. Each participant would fabricate its own hardware to execute the selected protocol components. Evaluation must connect the inspected design to the fabricated device, stating assumptions about fabrication, physical access, and device integrity. Synthesis is one stage of that investigation.

V. RESOURCES AND SUPPORT

We are seeking financial support, research collaborators, and access to computing and hardware prototyping resources to develop and demonstrate refereeless cryptographic protocols. Funding would support dedicated research time, independent cryptographic review, and implementation work. Access to FPGA boards and measurement equipment would enable the design and evaluation of hardware for participating parties.

We welcome collaboration in cryptography, game theory, formal methods, and hardware design. Our initial objective is one carefully specified and independently reviewed demonstrator, building on the finite-game work described here. Support would help turn this starting point into inspectable protocol specifications, reference implementations, adversarial tests, and measured software and hardware experiments, with the proof status and limits of each result stated openly.

Prospective supporters and collaborators can reach us at contact@waajacu.com to discuss funding, equipment access, or technical collaboration. The purpose is to make cooperation possible without requiring parties to surrender their legitimate secrets or strategic autonomy. Peace is the ethical motivation; it is not a property a cryptographic proof can establish.

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